KRUPA PATEL

Target Triage and Tractability Assessment for Parkinson's Disease

1. Re-ranking Analysis

The integration of the protein language model tractability prior with the Module 2 multi-evidence scores yielded a refined ranking for Parkinson's disease targets. The blending weight was set to $w = 0.6$, placing a 60% emphasis on the original Module 2 multi-evidence score percentile and a 40% emphasis on the best-modality tractability percentile.

The tractability prior introduced notable shifts compared to the raw Module 2 ranking:

- **FYN (+5 positions):** Moved up the most, rising from rank 14 to rank 9. This upward movement is driven by its exceptional small-molecule tractability probability ($p_{\text{sm}} = 0.96$) and high embedding similarity to established kinase drug targets like SRC (0.99) and LYN (0.97).
- **MAPT (−6 positions):** Dropped the most, falling from rank 9 to rank 15. Despite its established biological significance in tauopathies and Parkinson's disease, the model identifies very few small-molecule or antibody clinical precedents in its neighborhood ($p_{\text{sm}} = 0.01$, $p_{\text{ab}} = 0.00$), penalizing its tractability score.
- **LMNA and MAPT (Percentile-Noise and Prior Effects):** LMNA initially held the top position (rank 1) based on Module 2 evidence but shifted to rank 4 after incorporating tractability. Because LMNA lacks clinical precedence in the training panel ($p_{\text{sm}} = 0.00$, $p_{\text{ab}} = 0.00$), the low tractability prior pulls its final rank downward. Similarly, percentile-noise effects across distribution boundaries can alter rank positions when blending multi-evidence scores with neural embedding classifiers.

2. Modality Comparison

Comparing small-molecule (p_{sm}) and antibody (p_{ab}) tractability probabilities highlights how protein localization and functional class dictate therapeutic modality:

- **Cell-Surface Proteins vs. Kinases:** Cell-surface targets such as CHL1 ($p_{\text{sm}} = 0.48$, $p_{\text{ab}} = 0.10$) and PAM ($p_{\text{sm}} = 0.16$, $p_{\text{ab}} = 0.11$) exhibit moderate scores across both modalities, reflecting their membrane-bound architecture situated near receptors like ITGA2, NRP1, and IL1R1. In contrast, intracellular kinases like FYN ($p_{\text{sm}} = 0.96$, $p_{\text{ab}} = 0.01$) show high small-molecule tractability but near-zero antibody tractability. This is because FYN possesses an intracellular catalytic ATP-binding pocket accessible to small-molecule inhibitors, but lacks extracellular domains accessible to circulating therapeutic antibodies.

- **SNCA and MAPT:** Classical Parkinson's targets like SNCA (α-synuclein, $p_{\text{sm}} = 0.02$, $p_{\text{ab}} = 0.01$) and MAPT ($p_{\text{sm}} = 0.01$, $p_{\text{ab}} = 0.00$) yield low baseline clinical probabilities near zero, even though anti-α-synuclein antibodies are actively evaluated in clinical trials. This disparity occurs because historical training labels in clinical precedent datasets heavily favor traditional druggable classes (enzymes, receptors), leaving intrinsically disordered proteins or emerging antibody targets unrepresented in historical clinical approvals.

3. Validation and Generalization

Evaluating the logistic regression prior on the training panel provides rigorous performance metrics:

- **Performance Metrics:** The random-split cross-validation achieved an **AUROC of 0.848** and an **Average Precision (AP) of 0.731** (against a random chance baseline of 0.334 based on positive class prevalence). The more rigorous **family-split cross-validation yielded an AUROC of 0.754 and an Average Precision of 0.555**.

- **Why Family Split is the Honest Number:** Protein language model embeddings capture deep evolutionary and structural homology within protein families. A random split allows homologous proteins from the same structural family to intermix between training and validation folds, causing data leakage and overly optimistic metrics due to structural memorization. By utilizing a GroupKFold split across protein families (family_group), the model is forced to generalize to entirely unseen protein folds. This yields an honest generalization metric with typical ± 0.02 fold variability, reflecting real-world predictive utility on novel targets.

4. Variant Scoring Interpretation

Analysis of the anchor variant scores reveals important principles regarding protein language model representations:

- **Anchor Variant Scores:** APOE C130R ($\epsilon 4$, risk allele) scored $+7.21$ (100th percentile within the protein), whereas the protective APOE R176C ($\epsilon 2$) scored -8.56 (11.8th percentile). TREM2 R47H (Alzheimer's risk variant) scored -5.16 (47.7th percentile), and the TP53 R175H cancer hotspot scored -5.97 (19.3th percentile).

- **Pathogenicity vs. Evolutionary Score:** A strongly negative score does not mean "pathogenic" in a clinical disease context; rather, it indicates that a mutation is **evolutionarily unfavorable or disruptive** to the wild-type sequence context learned by the language model across millions of sequences.

- **Why Percentiles Beat Raw Scores:** Raw scores vary widely across different protein lengths and baseline distributions (e.g., comparing TREM2 directly to p53 is uninformative). Within-protein percentiles normalize these distributions, allowing consistent interpretation of mutational severity. Crucially, these sequence-based models **cannot see** 3D structural context, post-translational modifications, lipid/protein interactions, or non-coding regulatory regions—they evaluate statistical plausibility solely based on primary amino acid sequence co-evolution.

5. Limitations and Next Steps

- **What Sequence Alone Cannot Tell You:** Sequence embeddings capture evolutionary co-evolution and biochemical similarity, but they cannot determine 3D pocket druggability, binding kinetics, conformational flexibility, allosteric regulation, or the exact spatial orientation of catalytic residues.
- **Week 4 Structure-Based Pocket Assessment Picks:**
    1. **MAP4K4 (Rank 1, final score 0.827, $p_{\mathrm{sm}} = 0.67$):** Selected because of its exceptional small-molecule tractability score and well-defined kinase pocket architecture, making it an ideal candidate for structural pocket assessment and inhibitor docking.
    2. **PBXIP1 (Rank 2, final score 0.773, $p_{\mathrm{ab}} = 0.04$):** Selected to evaluate alternative modality interfaces and structural binding properties alongside its high composite triage rank.

Unmapped Genes:

None (0 unmapped genes out of 15 top targets). All selected targets were successfully mapped to proteome.tsv and proteome_emb_main.npy. If any non-coding or unreviewed genes had been encountered, they would have required genomic, transcriptomic, or literature-based evidence instead of protein language model embeddings.